PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lecture 5: Multi-Objective Loss and Covariance

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Overview

Fit quality, validation failures, model complexity, and compute cost are treated as separate objectives.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$L_R = r^T C^{-1} r$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$score(c) = (L_R, N_{legacyfail}, K, C_{compute})$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$Var_{total} = C_{data} + C_{num} + C_{loop}$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 5
- Multi-Objective Acceptance MOA-6
- Regression Barrier Guide